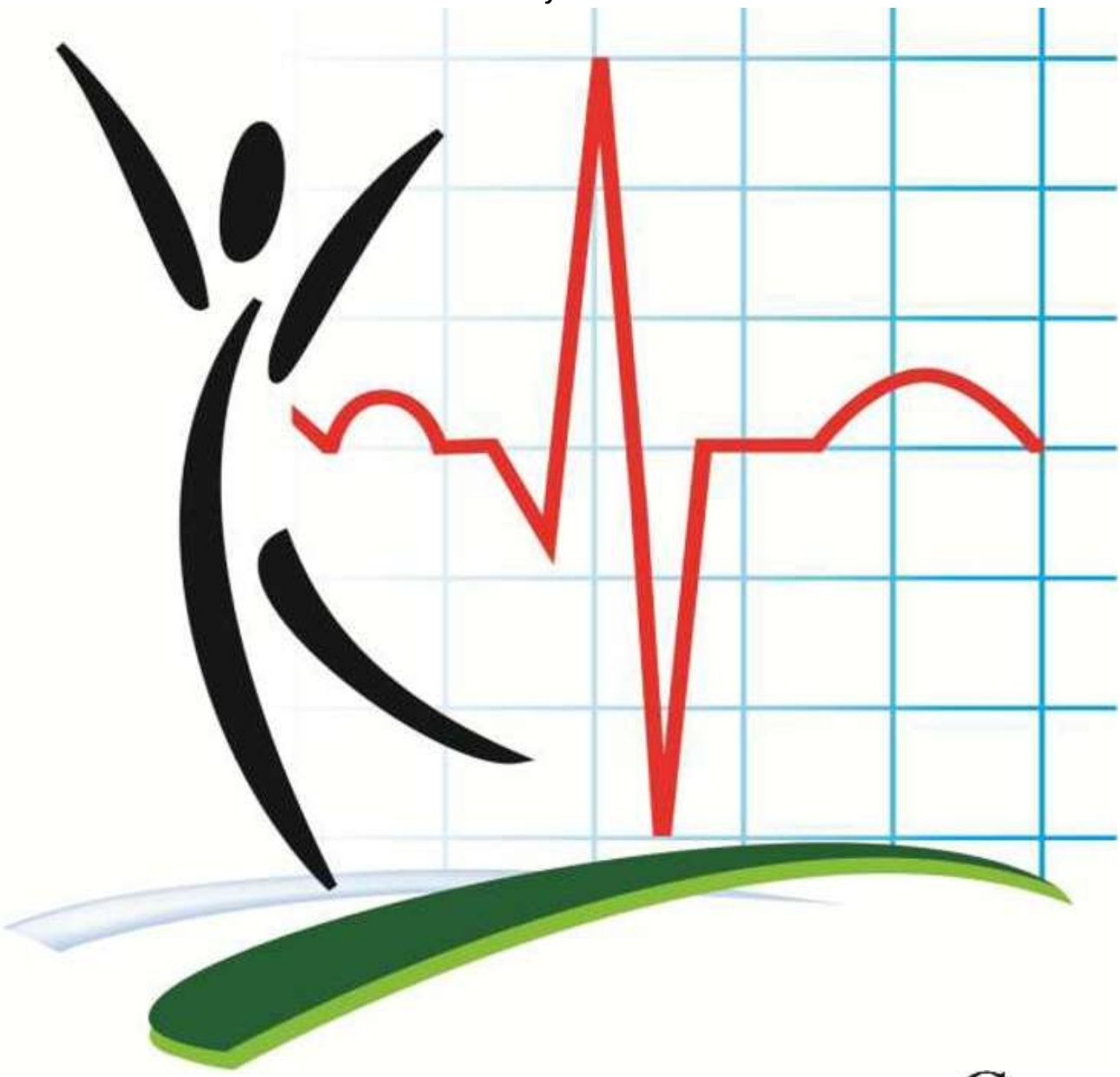
Canadian Community Health Survey (CCHS)

Annual component

Complement to the user guide

2015-2016 Public Use Microdata file

January 2019



Statistique Canada Statistics Canada

Canada

The following document is a complement to the 2015-2016 Canadian Community Health Survey (CCHS) Microdata User Guide. It presents the approach used to transform the 2015-2016 Master files into Public Use Microdata Files (PUMF) and lists which health regions have been grouped. Note that CCHS PUMFs are only comprised of the content of the master file and excludes any Rapid Response content, sub-samples, or sample buy-ins. For more information on these types of content, see the main CCHS user guide.

1. Documentation

The 2015-2016 questionnaire serves as reference for the 2015-2016 PUMF data file. The questionnaire will display additional questions that are not represented in the PUMF due to suppressions and other restriction methods that have been applied to the data.

The 2015-2016 PUMF data dictionary does not include the corresponding question names which can be found in the questionnaire. For variables that have been grouped or capped, the category 'Not Stated' includes the responses 'Don't Know', 'Refusal' and 'Not Stated'.

2. Conversion of 2015-2016 Master File to Public Use Microdata File (PUMF)

The approach for creating a PUMF is intended to balance the requirements for maintaining respondent confidentiality by minimizing disclosure risks, while providing the most useful data to users at the health region level. The following paragraphs outline some of the procedures applied to convert the CCHS master file into a PUMF.

Health regions: Health regions fall under provincial jurisdiction. As mentioned in Section 5.2 of the User Guide, there are 108 health regions across Canada. Thus, the master data file provides representative information for 108 HRs.

Some health regions have population sizes that were considered to be too small to appear individually in the PUMFs. These health regions were grouped with other(s) for the purpose of publication. The approach for keeping or collapsing health region geography in the PUMF entails applying a minimum population size for the given geography – 70, 000 inhabitants. This resulted in:

i) collapsing 19 HRs in 9 health region groupings in Newfoundland and Labrador, New Brunswick, Ontario, and Saskatchewan;

ii) one Quebec HR (Nord-du-Québec) being excluded from the PUMF due to its small population size and its demographic characteristics that did not lend itself well to being collapsed with other HRs;

iii) all sample design variables being excluded.

After the grouping of some HR's, the CCHS PUMF is comprised of 99 HRs/HR groupings across the country. The list of HR's before and after collapsing is available in section 5 of this document.

Optional content suppression: Due to the grouping of some HRs, optional content that was not common to each health region within the grouping would have been suppressed. Since 2005, the regions and provinces have opted to coordinate the optional content selected in order to ensure a uniform selection of optional modules provincially; therefore this suppression was not necessary for any provinces.

The 2015-2016 master and PUMF files only contain optional content data for modules that were asked in both years, which applies to the vast majority of optional content data.

Disclosure control: As mentioned earlier, the 2015-2016 PUMF is designed to preserve the analytical value of data as much as possible while ensuring the potential for identifying individual respondents is minimal. Several controls were implemented in creating the PUMF file. Restriction methods such as removing direct identifiers (e.g., sample ID, name, telephone number), reducing, recoding, and/or suppressing detail based on small frequencies given specific socio-demographic characteristics were used. Examples of variables from the master file that were not included in the PUMF due to a high risk of disclosure (either because the variable is a risk on its own or is so in combination with other variables) include: attempted suicide in the past 12 months, used ecstasy in lifetime, and whether the respondent is presently pregnant.

In addition, some response categories were deemed to be at possible risk of disclosure and were regrouped so as to be included on the PUMF. Examples include: body part affected by most serious injury, number of times consulting health professionals, number of years since stopping smoking, and main source of household income. With these regroupings in place, the 2015-2016 PUMF contains both fewer variables and is less detailed compared with the master file, while still preserving the analytical value of the data.

Age of respondent: Respondents' age is provided as age grouping in the CCHS PUMFs. Most of these groupings are found in 5-year age bands, from age 20-24 to age 75-79, with all respondents aged 80 and over comprising a single group. Prior to the CCHS 2005 PUMF, youth age groups included those aged 12-14 and 15-19. Beginning with the CCHS 2005 PUMF, the older youth age group has been split between those aged 15-17 and those aged 18-19. As a result, these groupings have three important benefits:

i) BMI data can be presented for those aged 12-17;

ii) data for those aged 18-19 is no longer automatically suppressed for variables that were only asked to those aged 18 and over.

Although information on some variables were collected for people of a specific age bracket, some data for certain ages still had to be suppressed to ensure confidentiality. For example, labour force data were collected for respondents aged 15 to 75. Because there is an age group 75-79 on the PUMF, publishing labour force data for 75-year olds would reveal their exact age by default. Therefore, labour force data (LBS) and some waiting times data (WTM) are only available on the PUMFs for respondents aged 15-74. Similar suppression was done for education (EDU) data of 12-14 year olds and maternal experience (MEX, MXS, and MXA) data for 50-year old women.

Special suppression was done for the three modules on maternal experiences (MEX, MXS, and MXA), data for women aged 15-19 because there was concern for a high risk of disclosure for women in the age group. As such, though MEX, MXS, and MXA data on the master file are shown for all women aged 15 to 55, the PUMF only present's data for women aged 20 to 49.

3. Bootstrap Weights for the PUMF

Previous CCHS PUMFs did not include bootstrap weights in part due to confidentiality concerns. The 2015-2016 PUMF includes bootstrap weights which are disseminated for users to estimate variances. For confidentiality reasons, the bootstrap weights for the public-use-microdata-file (PUMF) are not generated using the same methodology as for the CCHS master and share files. All bootstrap weights on the PUMF are strictly positive.

Also note that due to the removal of one health region from the PUMF, the sums of bootstrap weights by replicate are not exactly equal. The effect on the variance is negligible.

4. Content

The CCHS content can be divided into two categories, optional content and common content modules.

Optional content

The optional content component is designed to give provinces and territories the opportunity to select content that addresses their own public health priorities. The 2015-2016 PUMF includes only those modules that were selected for both years by a PUMF health region.

Common content

As mentioned in sections 4.1 and 4.2 of the User Guide, common content modules include questions asked of all respondents. This common content is divided into three components: the annual core content, and one year and two-year theme content.

5. Health Region Groupings

The 2015-2016 Master file 108 health regions have been grouped into 99 health regions for the PUMF. These groupings are detailed below. The PUMF health regions (variable GEODGHR4) are similar to what was produced for previous CCHS years.

Health regions for dissemination, before PUMF		Health regions for PUMF	
Newfoundland and Labrador		**Newfoundland and Labrador**	
1011	Eastern Regional Integrated Health Authority	10911	Eastern Regional Integrated Health Authority
1012	Central Regional Integrated Health Authority	10912	Central Regional Integrated Health Authority
1013	Western Regional Integrated Health Authority	10913	Western Region/Labrador-Grenfell Regional Integrated HA
1014	Labrador-Grenfell Regional Integrated Health Authority		
Prince Edward Island		**Prince Edward Island**	
1100	Prince Edward Island	11901	Prince Edward Island
Nova Scotia		**Nova Scotia**	
1201	Zone 1 - Western	12901	Zone 1 - Western
1202	Zone 2 - Northern	12902	Zone 2 - Northern
1203	Zone 3 - Eastern	12903	Zone 3 - Eastern
1204	Zone 4 - Central	12904	Zone 4 - Central
New Brunswick		**New Brunswick**	
1301	Zone 1	13901	Zone 1
1302	Zone 2	13902	Zone 2
1303	Zone 3	13903	Zone 3
1304	Zone 4	13904	Zones 4 and 5
1305	Zone 5		
1306	Zone 6	13906	Zones 6 and 7
1307	Zone 7		
Quebec		**Quebec**	
2401	Région du Bas-Saint-Laurent	24901	Région du Bas-Saint-Laurent
2402	Région du Saguenay - Lac-Saint-Jean	24902	Région du Saguenay - Lac-Saint-Jean
2403	Région de Québec	24903	Région de Québec
2404	Région de la Mauricie et du Centre-du-Québec	24904	Région de la Mauricie et du Centre-du-Québec
2405	Région de l'Estrie	24905	Région de l'Estrie
2406	Région de Montréal	24906	Région de Montréal
2407	Région de l'Outaouais	24907	Région de l'Outaouais
2408	Région de l'Abitibi-Témiscamingue	24908	Région de l'Abitibi-Témiscamingue
2409	Région de la Côte-Nord	24909	Région de la Côte-Nord
2410	Région du Nord-du-Québec		
2411	Région de la Gaspésie - Îles-de-la-Madeleine	24911	Région de la Gaspésie - Îles-de-la-Madeleine
2412	Région de la Chaudière-Appalaches	24912	Région de la Chaudière-Appalaches
2413	Région de Laval	24913	Région de Laval
2414	Région de Lanaudière	24914	Région de Lanaudière
2415	Région des Laurentides	24915	Région des Laurentides
2416	Région de la Montérégie	24916	Région de la Montérégie

Health regions for dissemination, before PUMF

Ontario

3526	Algoma
3527	Brant
3530	Durham
3531	Elgin-St. Thomas
3533	Grey Bruce
3534	Haldimand-Norfolk
3535	Haliburton, Kawartha, Pine Ridge
3536	Halton
3537	Hamilton
3538	Hastings and Prince Edward Counties
3539	Huron
3554	Perth
3540	Chatham-Kent
3541	Kingston, Frontenac and Lennox and Addington
3542	Lambton
3543	Leeds, Grenville and Lanark
3544	Middlesex-London
3546	Niagara
3547	North Bay Parry Sound
3563	Timiskaming
3549	Northwestern
3551	Ottawa
3552	Oxford
3553	Peel
3555	Peterborough
3556	Porcupine
3557	Renfrew
3558	Eastern Ontario
3560	Simcoe Muskoka
3561	Sudbury
3562	Thunder Bay
3565	Waterloo
3566	Wellington-Dufferin-Guelph
3568	Windsor-Essex
3570	York
3595	City of Toronto

Manitoba

4601	Winnipeg Regional Health Authority
4602	Prairie Mountain Health
4603	Interlake-Eastern Regional Health Authority
4604	Northern Regional Health Authority
4605	Southern Health

Health regions for PUMF

Ontario

35926	Algoma
35927	Brant
35930	Durham
35931	Elgin-St. Thomas
35933	Grey Bruce
35934	Haldimand-Norfolk
35935	Haliburton, Kawartha, Pine Ridge
35936	Halton
35937	Hamilton
35938	Hastings and Prince Edward
35939	Huron/Perth
35940	Chatham-Kent
35941	Kingston, Frontenac and Lennox and Addington
35942	Lambton
35943	Leeds, Grenville and Lanark
35944	Middlesex-London
35946	Niagara
35947	North Bay Parry Sound/Timiskaming
35949	Northwestern
35951	Ottawa
35952	Oxford
35953	Peel
35955	Peterborough
35956	Porcupine
35957	Renfrew
35958	Eastern Ontario
35960	Simcoe Muskoka
35961	Sudbury
35962	Thunder Bay
35965	Waterloo
35966	Wellington-Dufferin-Guelph
35968	Windsor-Essex
35970	York
35995	City of Toronto

Manitoba

46901	Winnipeg Regional Health Authority
46902	Prairie Mountain Health
46903	Interlake-Eastern Regional Health Authority
46904	Northern Regional Health Authority
46905	Southern Health

Health regions for dissemination, before PUMF		Health regions for PUMF	
Saskatchewan		**Saskatchewan**	
4701	Sun Country RHA	47901	Sun Country/Five Hills/Cypress
4702	Five Hills RHA		
4703	Cypress RHA		
4704	Regina Qu'Appelle RHA	47904	Regina Qu'Appelle RHA
4705	Sunrise RHA	47905	Sunrise/Kelsey Trail
4708	Kelsey trail RHA		
4706	Saskatoon RHA	47906	Saskatoon RHA
4707	Heartland RHA	47907	Heartland/Prairie North
4710	Prairie North RHA		
4709	Prince Albert Parkland RHA	47909	Prince Albert Parkland/Mamawetan-Keewatin Yatthé-Athabaska
4714	Mamawetan/Keewatin/Athabasca RHA		
Alberta		**Alberta**	
4831	South Zone	48931	South Zone
4832	Calgary Zone	48932	Calgary Zone
4833	Central Zone	48933	Central Zone
4834	Edmonton Zone	48934	Edmonton Zone
4835	North Zone	48935	North Zone
British Columbia		**British Columbia**	
5911	East Kootenay	59911	East Kootenay
5912	Kootenay-Boundary	59912	Kootenay-Boundary
5913	Okanagan	59913	Okanagan
5914	Thompson/Cariboo	59914	Thompson/Cariboo
5921	Fraser East	59921	Fraser East
5922	Fraser North	59922	Fraser North
5923	Fraser South	59923	Fraser South
5931	Richmond	59931	Richmond
5932	Vancouver	59932	Vancouver
5933	North Shore/Coast Garibaldi	59933	North Shore/Coast Garibaldi
5941	South Vancouver Island	59941	South Vancouver Island
5942	Central Vancouver Island	59942	Central Vancouver Island
5943	North Vancouver Island	59943	North Vancouver Island
5951	Northwest	59951	Northwest
5952	Northern Interior	59952	Northern Interior
5953	Northeast	59953	Northeast
Territories		**Territories**	
6001	Yukon Territory	60901	Yukon Territory
6101	Northwest Territories	61901	Northwest Territories
6201	Nunavut	62901	Nunavut